

● MEDIUM

● ADVANCED MATH

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# Nonlinear functions

10 questions · ~15 min

# 04

**Q1**

ADVANCED MATH

On April 1, there were 233 views of an advertisement posted on a website. Every 2 days after April 1, the number of views of the advertisement had increased by 70% of the number of views 2 days earlier. The function  $f$  gives the predicted number of views  $x$  days after April 1. Which equation defines  $f$ ?

- A.  $f(x) = 2330.70^{\frac{x}{2}}$
- B.  $f(x) = 2330.70^{2x}$
- C.  $f(x) = 2331.70^{\frac{x}{2}}$
- D.  $f(x) = 2331.70^{2x}$

$$f(x) = (x + 6)(x + 5)(x - 4)$$

The function  $f$  is given. Which table of values represents  $y = f(x) - 3$ ?

A.

| $x$ | $y$ |
|-----|-----|
| -6  | -9  |
| -5  | -8  |
| 4   | 1   |

B.

| $x$ | $y$ |
|-----|-----|
| -6  | -3  |
| -5  | -3  |
| 4   | -3  |

C.

| $x$ | $y$ |
|-----|-----|
| -6  | -3  |
| -5  | -2  |
| 4   | 7   |

D.

| $x$ | $y$ |
|-----|-----|
| -6  | 3   |
| -5  | 3   |
| 4   | 3   |

$$N(d) = 115(0.90)^d$$

The function  $N$  defined above can be used to model the number of species of brachiopods at various ocean depths  $d$ , where  $d$  is in hundreds of meters. Which of the following does the model predict?

- A. For every increase in depth by 1 meter, the number of brachiopod species decreases by 115.
- B. For every increase in depth by 1 meter, the number of brachiopod species decreases by 10%.
- C. For every increase in depth by 100 meters, the number of brachiopod species decreases by 115.
- D. For every increase in depth by 100 meters, the number of brachiopod species decreases by 10%.

$$y = 0.25x^2 - 7.5x + 90.25$$

The equation gives the estimated stock price  $y$ , in dollars, for a certain company  $x$  days after a new product launched, where  $0 \leq x \leq 20$ . Which statement is the best interpretation of  $x, y = 1, 83$  in this context?

- A. The company's estimated stock price increased \$ 83 every day after the new product launched.
- B. The company's estimated stock price increased \$ 1 every 83 days after the new product launched.
- C. 1 day after the new product launched, the company's estimated stock price is \$ 83 .
- D. 83 days after the new product launched, the company's estimated stock price is \$ 1.

A scientist initially measures 12,000 bacteria in a growth medium. 4 hours later, the scientist measures 24,000 bacteria. Assuming exponential growth, the formula  $P = C2^{rt}$  gives the number of bacteria in the growth medium, where  $r$  and  $C$  are constants and  $P$  is the number of bacteria  $t$  hours after the initial measurement. What is the value of  $r$ ?

- A.  $\frac{1}{12,000}$
- B.  $\frac{1}{4}$
- C. 4
- D. 12,000

For the exponential function  $f$ , the value of  $f(0)$  is  $c$ , where  $c$  is a constant. Of the following equations that define the function  $f$ , which equation shows the value of  $c$  as the coefficient or the base?

A.  $f(x) = 221.5^{x+1}$

B.  $f(x) = 331.5^x$

C.  $f(x) = 49.51.5^{x-1}$

D.  $f(x) = 74.251.5^{x-2}$

$$f(x) = x^3 + 3x^2 - 6x - 1$$

For the function  $f$  defined above, what is the value of  $f(-1)$ ?

- A.  $-11$
- B.  $-7$
- C.  $7$
- D.  $11$



A function  $p$  estimates that there were 2,000 animals in a population in 1998. Each year from 1998 to 2010, the function estimates that the number of animals in this population increased by 3% of the number of animals in the population the previous year. Which equation defines this function, where  $px$  is the estimated number of animals in the population  $x$  years after 1998?

- A.  $px = 2,0003^x$
- B.  $px = 2,0001.97^x$
- C.  $px = 2,0001.03^x$
- D.  $px = 2,0000.97^x$

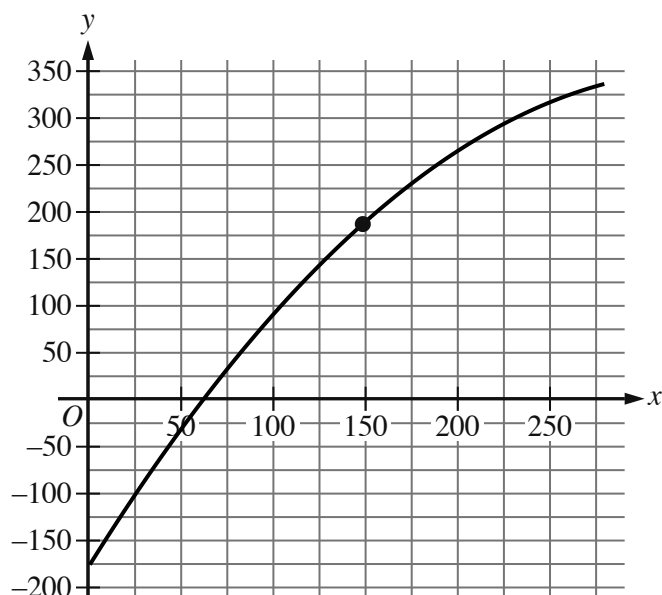
The number of bacteria in a liquid medium doubles every day. There are 44,000 bacteria in the liquid medium at the start of an observation. Which represents the number of bacteria,  $y$ , in the liquid medium  $t$  days after the start of the observation?

A.  $y = \frac{1}{2}44,000^t$

B.  $y = 244,000^t$

C.  $y = 44,000\frac{1}{2}^t$

D.  $y = 44,0002^t$



- The parabola opens downward.
- The parabola passes through the following approximate points:
  - (50 comma negative 33.83)
  - (149.02 comma 186.05)
  - (250 comma 315.37)

The graph shows the estimated boiling point  $y$ , in degrees Celsius, of a normal paraffin with a molecular weight of  $x$  grams per mole, where  $1 \leq x \leq 280$ . Which statement is the best interpretation of the point 149.02186.05?

- A. A normal paraffin with a molecular weight of 186.05 grams per mole has an estimated boiling point of 149.02 degrees Celsius.
- B. A normal paraffin with a molecular weight of 149.02 grams per mole has an estimated boiling point of 186.05 degrees Celsius.
- C. The minimum estimated boiling point for normal paraffins corresponds to a paraffin with a molecular weight of 149.02 grams per mole and an estimated boiling point of 186.05 degrees Celsius.

- D. The maximum estimated boiling point for normal paraffins corresponds to a paraffin with a molecular weight of 149.02 grams per mole and an estimated boiling point of 186.05 degrees Celsius.